

| Assessment | Test                | Domain                | Skill                     | Difficulty                               |
|------------|---------------------|-----------------------|---------------------------|------------------------------------------|
| SAT        | Reading and Writing | Information and Ideas | Central Ideas and Details | Hard <div></div> <div></div> <div></div> |

Question

For centuries, the widespread acknowledgment of the involvement of the cerebellum—a dense brain structure in vertebrates—in coordinating motor control in humans has hindered recognition of other possible functions of the structure. Neuroscience research from the last two decades now suggests that the cerebellum regulates emotion and social behavior, and recent research by Ilaria Carta and colleagues has identified a pathway connecting the cerebellum to a center for motivation and reward processing known as the ventral tegmental area (VTA).

Which choice best states the main idea of the text?

Answer

- A. The recent verification of a pathway between the VTA and the cerebellum confirms the cerebellum’s long-suspected role in motor coordination.
- B. Recent advances in the field of neuroscience have challenged widely accepted claims about the function of a pathway connecting the VTA and the cerebellum.
- C. The cerebellum has primarily been thought to regulate motor functioning, but in recent years neuroscience researchers have been uncovering additional functions.
- D. Technological limitations have historically hindered the study of the cerebellum, but the recent development of new technologies has led to greater insights into its functions.

Correct Answer: C

Rationale

Choice C is the best answer. The text says that the cerebellum has long been thought to regulate motor functioning, but new research shows that it may also have other functions—including regulating emotion and social behavior and some connection to motivation and rewards processing.

Choice A is incorrect. The VTA is described as “a center for motivation and reward processing,” and the discovery of the pathway between the VTA and the cerebellum supports the theory that the cerebellum is involved in functions other than motor coordination. Choice B is incorrect. The text says that recent research has identified this pathway, but it doesn’t discuss any previous “widely accepted claims” about the pathway’s function. The “widespread acknowledgement” mentioned early in the passage is about the cerebellum alone, not its connection to the VTA. Choice D is incorrect. The text never discusses any technological limitations or any new technologies.